

3 by 3 eigenvalues/eigenvectors example

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Find the eigenvalues and eigenvectors of the matrix

$$A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

We have

$$A - \lambda I = \begin{bmatrix} -\lambda & 1 & 1 \\ 1 & -\lambda & 1 \\ 1 & 1 & -\lambda \end{bmatrix}$$

Using cofactor expansion along the first row, we see that the determinant of $A - \lambda I$ is

$$\begin{aligned} -\lambda(\lambda^2 - 1) - 1(-\lambda - 1) + 1(1 + \lambda) &= -\lambda^3 + \lambda + \lambda + 1 + 1 + \lambda \\ &= -\lambda^3 + 3\lambda + 2 \end{aligned}$$

Now we have to find the roots of this polynomial. To narrow down the possibilities for the roots, it helps to use the Rational Root Theorem:

Theorem. Suppose that

$$p(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$$

is a polynomial with *integer* coefficients. Then each rational zero of $p(x)$ must have the form p/q , where

- p is a factor of a_0 , and
- q is a factor of a_n .

The theorem is not that hard to prove. Basically, you plug in $x = p/q$ into the polynomial and get 0, and then use some facts about divisibility.

We will apply the Rational Roots Theorem to the polynomial $-x^3 + 3x + 2$. By the Rational Roots Theorem, the only rational roots of $-x^3 + 3x + 2$ must have the form p/q , where p is a factor

of -1 and q is a factor of 2 (keep in mind that you also have to consider negative factors). This means that the only possible rational roots of $-x^3 + 3x + 2$ are

$$1, \quad -1, \quad 1/2, \quad \text{and} \quad -1/2.$$

Now we go one by one and test the possible values to see if they are roots. Testing 1, we get

$$-(1)^3 + 3(1) + 2 = 4,$$

so 1 is not a root. What about -1 ? We get

$$-(-1)^3 + 3(-1) + 2 = 1 - 3 + 2 = 0,$$

so -1 is a root. Therefore, $x - (-1)$ (which is $x + 1$) is a factor of $-x^3 + 3x + 2$. To find the quotient, we use long division (the steps are omitted) to get

$$-x^3 + 3x + 2 = (x+1)(-x^2 + x + 2) = -(x+1)(x^2 - x - 2) = -(x+1)(x+1)(x-2) = -(x+1)^2(x-2)$$

So the roots of $-x^3 + 3x + 2$ are -1 (with multiplicity 2) and 2 (with multiplicity 1). These are the eigenvalues of A .

Now we have to find a basis for the eigenspaces. For $\lambda_1 = -1$, we calculate $A - \lambda I$ and bring it to row echelon form to solve the homogeneous equation $(A - \lambda I)\mathbf{x} = \mathbf{0}$:

$$A - \lambda_1 I = A + I = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

We need two independent solutions to $x + y + z = 0$. The easiest way is to just pick numbers that work instead of systematically doing it:

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}$$

For $\lambda_2 = 2$, we get

$$\begin{aligned} A - \lambda_2 I = A - 2I &= \begin{bmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{bmatrix} \rightarrow \begin{bmatrix} -2 & 1 & 1 \\ 0 & -3/2 & 3/2 \\ 0 & 3/2 & -3/2 \end{bmatrix} \rightarrow \begin{bmatrix} -2 & 1 & 1 \\ 0 & -3/2 & 3/2 \\ 0 & 0 & 0 \end{bmatrix} \\ &\rightarrow \begin{bmatrix} 1 & -1/2 & -1/2 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix} \end{aligned}$$

The solutions $\mathbf{x} = (x, y, z)$ to $(A - 2I)\mathbf{x} = \mathbf{0}$ are the vectors satisfying $x = z$, $y = z$, so one solution is

$$\mathbf{v}_3 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

So the -1 -eigenspace is the set of all vectors of the form

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2$$

for constants c_1 and c_2 , and the 2 -eigenspace is the set of all vectors of the form

$$c_3\mathbf{v}_3$$

for constants c_3 .